



COMSATS University Islamabad

Department of Computer Science

Course Information

Course Code: **CSC354**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Machine Learning**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

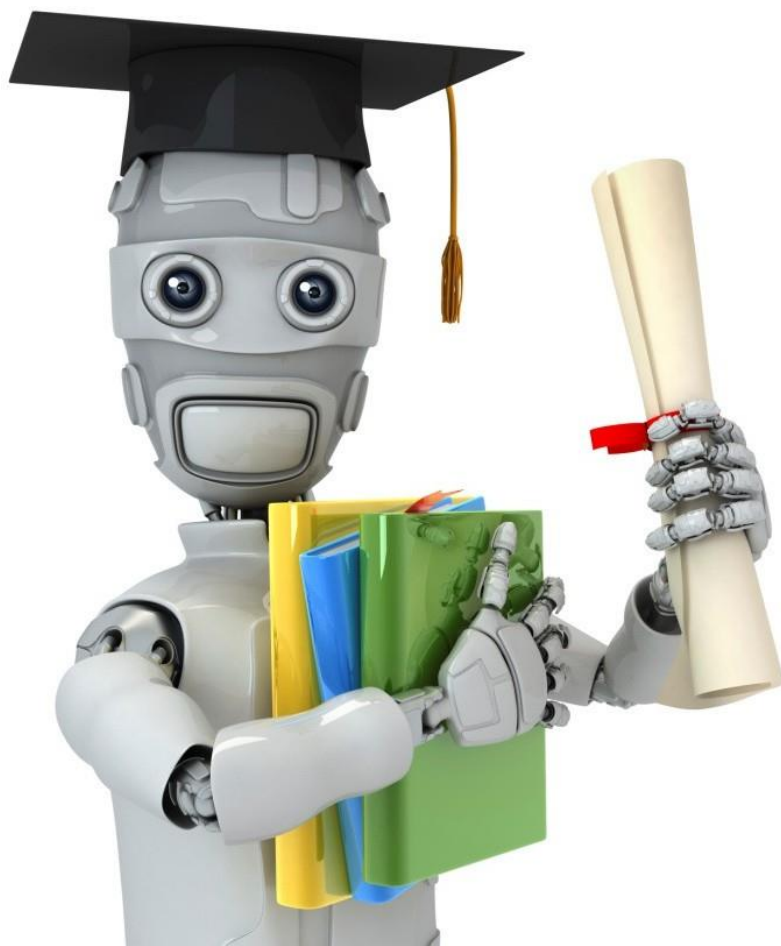
Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O'Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.



COMSATS University Islamabad
Department of Computer Science

- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on zeenatzulfiqar@cuiatd.edu.pk



Machine Learning

Introduction

Unsupervised Learning Model (Agglomerative Clustering)

Agglomerative Clustering

- **Agglomerative Clustering** is a bottom-up hierarchical clustering algorithm in which each data point starts as its own cluster, and the closest clusters are merged step by step until the required number of clusters is formed.
- **Type:** Hierarchical clustering **Approach:** Bottom-up **Initial State:** Each data point is a separate cluster. **Process:** Merge the two nearest clusters repeatedly. **Stopping Condition:** Continue until one cluster remains or a specified number of clusters is reached.

When Should We Use Agglomerative Clustering?

- Agglomerative Clustering is used for **unsupervised learning problems**, where the data has **no class labels** and the goal is to discover natural groups (clusters) in the data.
- **Suitable Problems**
- **Customer Segmentation:** Group customers based on purchasing behavior.
- **Document Clustering:** Organize similar documents or news articles.
- **Image Segmentation:** Divide an image into meaningful regions.
- **Gene Expression Analysis:** Group genes with similar expression patterns.
- **Social Network Analysis:** Identify communities of users with similar connections.
- **Recommendation Systems:** Group users or products with similar characteristics.
- **Anomaly Detection:** Identify data points that do not belong to any cluster.

When Should We Use Agglomerative Clustering?

When is it Preferred?

- Use Agglomerative Clustering when:
- The dataset is **small to medium-sized**.
- You **do not know the number of clusters** in advance.
- You want to **visualize the hierarchy of clusters** using a dendrogram.
- The data may have **irregularly shaped clusters**.

When Should We Use Agglomerative Clustering?

- **When Should You Avoid It?**
- Avoid using Agglomerative Clustering when:
- The dataset is **very large**, as it is computationally expensive.
- Fast clustering is required.
- The data contains many outliers or noisy points.

Working Process

- **How Agglomerative Clustering Works**
- Start with each data point as a separate cluster.
- Calculate the distance between all clusters.
- Find the two closest clusters.
- Merge these clusters.
- Recalculate distances.
- Repeat until the required number of clusters is formed.

Example 1

- **Question**

Consider the following **6 one-dimensional data points**:

- 18, 22, 25, 27, 42, 43
- Apply the **Agglomerative Hierarchical Clustering** algorithm using the **Single Linkage (Minimum Distance)** method.

Perform the following tasks:

1. Construct the initial proximity (distance) matrix.
2. Merge the two closest clusters in each iteration.
3. Update the proximity matrix after every merge.
4. Continue until all data points belong to a single cluster.
5. Draw the final dendrogram.

Example 1

Solution

Step 1: Initial Clusters

Initially, every data point is considered as a separate cluster.

Cluster

{18}

{22}

{25}

{27}

{42}

{43}

Step 2: Construct the Initial Proximity Matrix

Row 1 (Distances from 18)

Calculation

Distance

$|18-18|$

0

$|18-22|$

4

$|18-25|$

7

$|18-27|$

9

$|18-42|$

24

$|18-43|$

25

Step 2: Construct the Initial Proximity Matrix

Row 2 (Distances from 22)

Calculation	Distance
$ 22-18 $	4
$ 22-22 $	0
$ 22-25 $	3
$ 22-27 $	5
$ 22-42 $	20
$ 22-43 $	21

Step 2: Construct the Initial Proximity Matrix

Row 3 (Distances from 25)

Calculation	Distance
$ 25-18 $	7
$ 25-22 $	3
$ 25-25 $	0
$ 25-27 $	2
$ 25-42 $	17
$ 25-43 $	18

Step 2: Construct the Initial Proximity Matrix

Row 4 (Distances from 27)

Calculation	Distance
$ 27-18 $	9
$ 27-22 $	5
$ 27-25 $	2
$ 27-27 $	0
$ 27-42 $	15
$ 27-43 $	16

Step 2: Construct the Initial Proximity Matrix

Row 5 (Distances from 42)

Calculation	Distance
$ 42-18 $	24
$ 42-22 $	20
$ 42-25 $	17
$ 42-27 $	15
$ 42-42 $	0
$ 42-43 $	1

Step 2: Construct the Initial Proximity Matrix

Row 6 (Distances from 43)

Calculation	Distance
$ 43-18 $	25
$ 43-22 $	21
$ 43-25 $	18
$ 43-27 $	16
$ 43-42 $	1
$ 43-43 $	0

Step 2: Construct the Initial Proximity Matrix

- Initial Proximity Matrix

	18	22	25	27	42	43
18	0	4	7	9	24	25
22	4	0	3	5	20	21
25	7	3	0	2	17	18
27	9	5	2	0	15	16
42	24	20	17	15	0	1
43	25	21	18	16	1	0

Iteration 1

- **Find the smallest non-zero distance**
- The minimum distance is **1** between **42** and **43**.

Update distances (Single Linkage)

- $\text{Distance}(\{42,43\},18)$
- $=\min(24,25)=24$
- $\text{Distance}(\{42,43\},22)$
- $=\min(20,21)=20$
- $\text{Distance}(\{42,43\},25)$
- $=\min(17,18)=17$
- $\text{Distance}(\{42,43\},27)$
- $=\min(15,16)=15$

	18	22	25	27	42,43
18	0	4	7	9	24
22	4	0	3	5	20
25	7	3	0	2	17
27	9	5	2	0	15
42,43	24	20	17	15	0

Iteration 2

- Smallest distance = **2**
- Merge:
- $\{25\} + \{27\} = \{25, 27\}$
- Current clusters:
- $\{18\}$ $\{22\}$ $\{25, 27\}$ $\{42, 43\}$
- **Update distances**
- $\text{Distance}(\{25, 27\}, 18)$
- $= \min(7, 9) = 7$
- $\text{Distance}(\{25, 27\}, 22)$
- $= \min(3, 5) = 3$
- $\text{Distance}(\{25, 27\}, \{42, 43\})$
- $= \min(17, 18, 15, 16) = 15$

	18	22	25,27	42,43
18	0	4	7	24
22	4	0	3	20
25,27	7	3	0	15
42,43	24	20	15	0

Iteration 3

- Smallest distance = **3**
- Merge
- $\{22\} + \{25, 27\} = \{22, 25, 27\}$
- Current clusters
- $\{18\}$ $\{22, 25, 27\}$ $\{42, 43\}$
- **Update distances**
- $\text{Distance}(\{22, 25, 27\}, 18)$
- $= \min(4, 7, 9) = 4$
- $\text{Distance}(\{22, 25, 27\}, \{42, 43\})$
- $= \min(20, 21, 17, 18, 15, 16) = 15$

	18	22,25,27	42,43
18	0	4	24
22,25,27	4	0	15
42,43	24	15	0

Iteration 4

- Smallest distance = 4
- Merge
- $\{18\} + \{22, 25, 27\} = \{18, 22, 25, 27\}$
- Current clusters
- $\{18, 22, 25, 27\}$ $\{42, 43\}$
- **Update Matrix**
- Distance
- $= \min(24, 25, 20, 21, 17, 18, 15, 16)$
- =15

	18,22,25,27	42,43
18,22,25,27	0	15
42,43	15	0

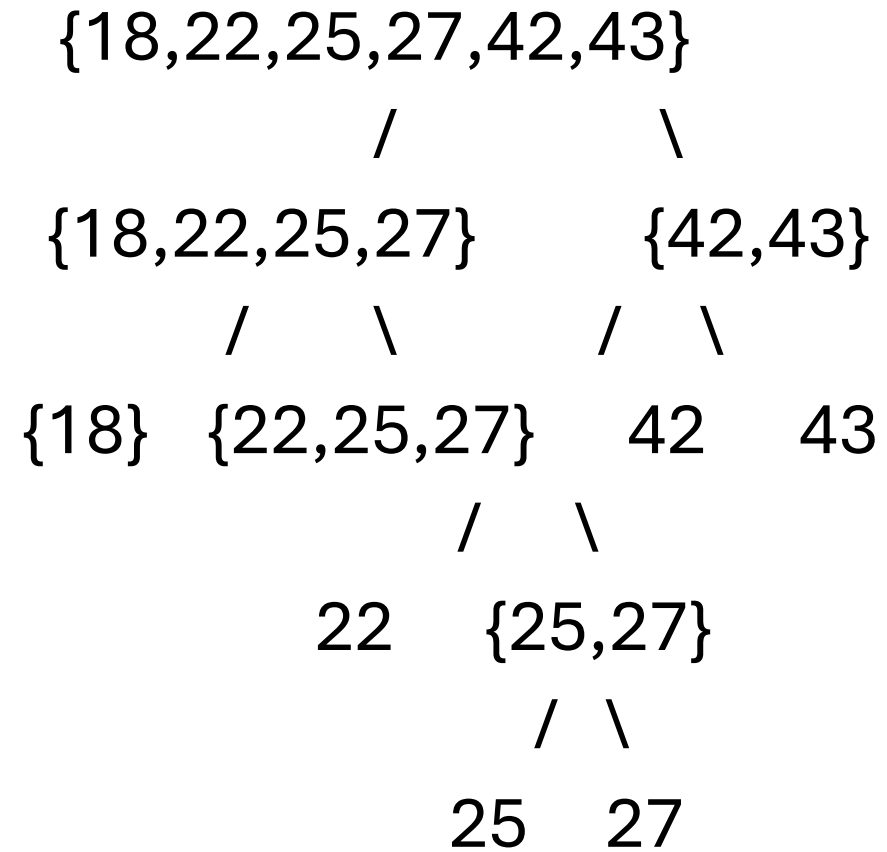
Iteration 5

- Smallest distance = **15**
- Merge
- $\{18, 22, 25, 27\} + \{42, 43\}$
=
 $\{18, 22, 25, 27, 42, 43\}$
- Only one cluster remains, so the algorithm stops.

Final Order of Merging

Step	Clusters Merged	Distance
1	{42} + {43}	1
2	{25} + {27}	2
3	{22} + {25,27}	3
4	{18} + {22,25,27}	4
5	{18,22,25,27} + {42,43}	15

Final Dendrogram



Agglomerative Clustering Solved Example 2

Solved Example 2 Cluster Using single link technique

Problem Definition:

For the given dataset find the clusters using a single link technique. Use Euclidean distance and draw the Dendrogram.

Sample No.	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Solved Example 2 Cluster Using single link technique

Step 1: Compute the distance matrix

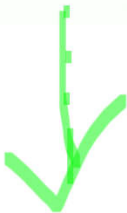
- So we have to find the Euclidean distance between each and every points.
- Let $A(x_1, y_1)$ and $B(x_2, y_2)$ are two points.
- Then Euclidean distance between

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Solved Example 2 Cluster Using single link technique

$$d(p_1, p_2) = \sqrt{(0.22 - 0.40)^2 + (0.38 - 0.53)^2} \\ = 0.23$$

$$d(p_1, p_3) = \sqrt{(0.35 - 0.40)^2 + (0.32 - 0.53)^2} \\ = 0.22$$

$$d(p_2, p_3) = \sqrt{(0.35 - 0.22)^2 + (0.32 - 0.38)^2} \\ = 0.14$$


Sample No.	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Solved Example 2 Cluster Using single link technique

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P1	0.40	0.53
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P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.14	0			
P4	0.37	0.19	0.13	0		
P5	0.34	0.14	0.28	0.23	0	
P6	0.24	0.24	0.10	0.22	0.39	0

Solved Example 2 Cluster Using single link technique

Step 2: Merging the two closest members.

- Here the **minimum value is 0.10** and hence we combine P3 and P6 (as 0.10 came in the P6 row and P3 column).
- Now, form clusters of elements corresponding to the minimum value and update the distance matrix.

Solved Example 2 Cluster Using single link technique

$$\begin{pmatrix}
 & P1 & P2 & P3 & P4 & P5 & P6 \\
 P1 & 0 & & & & & \\
 P2 & 0.23 & 0 & & & & \\
 P3 & 0.22 & 0.14 & 0 & & & \\
 P4 & 0.37 & 0.19 & 0.13 & 0 & & \\
 P5 & 0.34 & 0.14 & 0.28 & 0.23 & 0 & \\
 P6 & 0.24 & 0.24 & 0.10 & 0.22 & 0.39 & 0
 \end{pmatrix}$$

$$\begin{pmatrix}
 & P1 & P2 & P3, P6 & P4 & P5 \\
 P1 & 0 & & & & \\
 P2 & 0.23 & 0 & & & \\
 P3, P6 & 0.22 & 0.14 & 0 & & \\
 P4 & 0.37 & 0.19 & 0.13 & 0 & \\
 P5 & 0.34 & 0.14 & 0.28 & 0.23 & 0
 \end{pmatrix}$$

(P3, P6)

Solved Example 2 Cluster Using single link technique

$$\begin{pmatrix}
 & P1 & P2 & P3, P6 & P4 & P5 \\
 P1 & 0 & & & & \\
 P2 & 0.23 & 0 & & & \\
 P3, P6 & 0.22 & 0.14 & 0 & & \\
 P4 & 0.37 & 0.19 & 0.13 & 0 & \\
 P5 & 0.34 & 0.14 & 0.28 & 0.23 & 0
 \end{pmatrix}
 \quad
 \begin{pmatrix}
 & P1 & P2 & P3, P6, P4 & P5 \\
 P1 & 0 & & & \\
 P2 & 0.23 & 0 & & \\
 P3, P6, P4 & 0.22 & 0.14 & 0 & \\
 P5 & 0.34 & 0.14 & 0.28 & 0
 \end{pmatrix}$$

{(P3, P6), P4}

Solved Example 2 Cluster Using single link technique

$$\begin{pmatrix}
 & P1 & P2 & P3, P6, P4 & P5 \\
 P1 & 0 & & & \\
 P2 & 0.23 & 0 & & \\
 P3, P6, P4 & 0.22 & 0.14 & 0 & \\
 P5 & 0.34 & 0.14 & 0.28 & 0
 \end{pmatrix}
 \quad
 \begin{pmatrix}
 & P1 & P2, P5 & P3, P6, P4 \\
 P1 & 0 & & \\
 P2, P5 & 0.23 & 0 & \\
 P3, P6, P4 & 0.22 & 0.14 & 0
 \end{pmatrix}$$

{(P3, P6), P4} and **(P2, P5)**

Solved Example 2 Cluster Using single link technique

$$\begin{pmatrix} & P1 & P2, P5 & P3, P6, P4 \\ P1 & 0 & & \\ P2, P5 & 0.23 & 0 & \\ P3, P6, P4 & 0.22 & 0.14 & 0 \end{pmatrix}$$

$[\{(P3, P6), P4\}, (P2, P5)]$

$[\{(P3, P6), P4\}, (P2, P5)], P1$

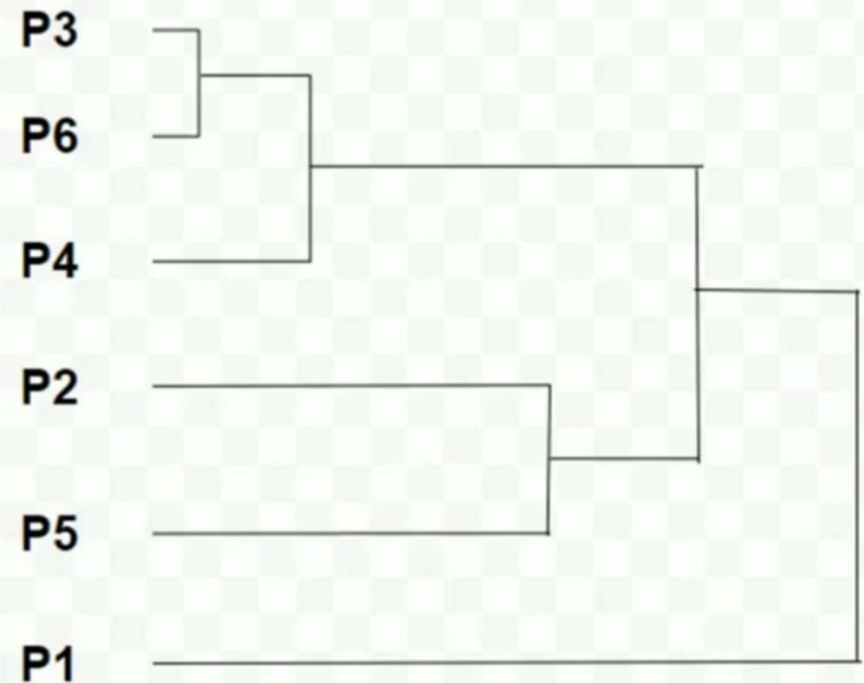
$(0.22 < 0.23)$

$$\begin{pmatrix} & P1 & P2, P5, P3, P6, P4 \\ P1 & 0 & \\ P2, P5, P3, P6, P4 & 0.22 & 0 \end{pmatrix}$$

Solved Example 2 Cluster Using single link technique

So now we have reached to the solution, the dendrogram for those question will be as follows:

[{(P3, P6), P4}, (P2, P5)], P1



Dendrogram of the cluster formed